

# Day 1: Foundations of Relativistic Kinematics & Experimental HEP

## 14-Day Hands-on Course on Heavy-Ion Phenomenology

Lecture

NIT-J,HEP-119

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# Syllabus: 14-Day Hands-on Course Layout

## Part I: Relativistic Kinematics

- Day 1: Experimental HEP, Boosts & Rapidity
- Day 2: Kinematics I: 4-vectors & CMS energy
- Day 3: Kinematics II: Rapidity & Pseudorapidity
- Day 4: Kinematics III: Invariant Yield & Phase Space

## Part II: AMPT Physics & Mechanics

- Day 5: AMPT Evolution: HIJING, ZPC, Hadronization, ART
- Day 6: AMPT Data format: event headers & parsing
- Day 7: Single-Particle Observables: Multiplicity &  $p_T$

## Part III: Thermodynamics & Collectivity

- Day 8: Transverse Spectra, Identified Particles &  $m_T$  Scaling
- Day 9: Centrality & Geometry: Glauber vs AMPT
- Day 10: Collective Flow: Elliptic Flow ( $v_2$ )
- Day 11: Medium Modification: ZPC Knobs & Mean Free Path

## Part IV: Advanced Scans & Pipeline

- Day 12: Energy Scan: Beam Energy Scan (STAR at RHIC)
- Day 13: Advanced Analysis
- Day 14: Wrap-Up: Reproducible HEP Data Pipeline

# Why Experimental High Energy Physics? Probing the Extremes

## The QGP Phase & Early Universe

- We collide heavy nuclei at ultra-relativistic velocities to recreate the conditions of the early universe ( $10^{-6}$  seconds after the Big Bang).
- **The Quark-Gluon Plasma (QGP):** A state of deconfined, color-charged quarks and gluons.

## The QCD Confinement Challenge

- Hadrons are held together by strong force (QCD). Because of color confinement, we cannot isolate individual quarks/gluons directly.
- **Solution:** We must analyze final-state hadron sprays to infer the plasma's thermodynamic and transport properties.

# Collision Energetics: Why Accelerators?

## Resolving Matter at Sub-Femtometer Scales

- Accelerators provide high center-of-mass energies ( $\sqrt{s}$ ) to satisfy mass-energy equivalence ( $E = mc^2$ ).
- Short de Broglie wavelengths probe smaller structures:  $\lambda = h/p$ .

## Center-of-Mass Energy Formulas

- **\*\*Collider Advantage:\*\*** In a collider ( $E_{beam1} = E_{beam2} = E$ ), center-of-mass energy scales linearly:

$$\sqrt{s_{\text{collider}}} \approx 2E$$

- **\*\*Fixed Target Limit:\*\*** In a fixed-target collision (beam energy  $E$  on target nucleon mass  $m_N$ ), energy scales only as the square root:

$$\sqrt{s_{\text{fixed}}} \approx \sqrt{2m_N E}$$

# Fixed-Target vs. Collider Geometry

## Center-of-Mass Energetics Comparison

Let us calculate the beam energy required to match LHC and RHIC center-of-mass energies in a fixed-target configuration:

| System             | Collider $\sqrt{s_{NN}}$ | Fixed-Target Equivalent $E_{\text{beam}}$ | Kinematic Advantage                |
|--------------------|--------------------------|-------------------------------------------|------------------------------------|
| SPS ( $p + p$ )    | 17.3 GeV                 | 158 GeV                                   | Baseline                           |
| RHIC ( $Au + Au$ ) | 200 GeV                  | 21.3 TeV                                  | 100× Beam boost                    |
| LHC ( $Pb + Pb$ )  | 5.02 TeV                 | 13.5 PeV (13,500 TeV)                     | Outside classical accelerator tech |

## Interactive Simulation

Open our custom simulator at [animations/collision\\_animation.html](animations/collision_animation.html) to interactively run collisions and observe the center-of-mass energy transformations.

# Specialization by Design: Belle II vs. LHCb

## Belle II (KEKB Asymmetric Collider)

- **Beams:** 8.0 GeV  $e^-$  on 3.5 GeV  $e^+$ .
- **Design:** Collides asymmetric energies to provide a longitudinal Lorentz boost to the center of mass.
- **Physics:** Allows precise decay-length measurements of short-lived B-mesons ( $B \rightarrow J/\psi K_S^0$ ) to study CP violation.

## LHCb (LHC Forward Spectrometer)

- **Beams:** Symmetric proton-proton at 13.6 TeV.
- **Design:** Forward single-arm cone spectrometer ( $2 < \eta < 5$ ).
- **Physics:** At LHC energies,  $b\bar{b}$  pairs are produced at very forward/backward angles. Placed in the forward direction to capture these heavy flavor hadrons.

# Specialization by Design: STAR vs. CMS vs. ALICE

## ALICE (LHC)

### QGP Specialized Tracking

- Optimized for extreme track density ( $Pb + Pb$  collisions).
- Extremely low material budget and magnetic field (0.5 T) to track and identify very low- $p_T$  hadrons.

## STAR (RHIC)

### Hadron Spectroscopy

- Massive Time Projection Chamber (TPC) optimized for particle identification, centrality scans, and collective flow physics at RHIC energies.

## CMS (LHC)

### High- $p_T$ & High-Luminosity

- Built around a massive 3.8 T superconducting solenoid.
- High-granularity silicon trackers and deep EM/Hadronic calorimeters designed to capture high- $p_T$  jets, leptons, and Higgs candidates.

# Layered "Onion" Detector Architecture

## Concentric Cylindrical Configurations

To track and reconstruct every single final-state particle, modern cylindrical colliders use concentric detector layers centered around the beam axis:

- ❶ **\*\*Vertex Detectors (Inner Silicon):\*\*** Capture prompt secondary decays close to the interaction point.
- ❷ **\*\*Central Trackers (TPC / Drift Chambers):\*\*** Map trajectories under magnetic fields to resolve momenta.
- ❸ **\*\*Particle Identification (TOF / RICH):\*\*** Separate pions, kaons, and protons.
- ❹ **\*\*Electromagnetic Calorimeters (EMCal):\*\*** Capture electron and photon energy.
- ❺ **\*\*Hadronic Calorimeters (HCal):\*\*** Absorb hadronic shower energy.
- ❻ **\*\*Muon Spectrometer:\*\*** Outermost chambers that capture highly penetrating muons.

# Tracking & Curvature in Time Projection Chambers (TPC)

## TPC Operation Principles

- Gas-filled drift chamber placed inside high electric and magnetic fields.
- Traversing charged particles ionize gas atoms, producing electrons that drift towards anodes.
- Reconstructs 3D tracks from anode pixels ( $X, Y$ ) and drift times ( $Z$ ).

## TPC Physics Outputs

- **Momentum Measurement:** Curved tracks in magnetic field  $B$  yield momentum  $p$ :

$$p = 0.3 \cdot B \cdot R$$

where  $R$  is the curvature radius.

- **Energy Loss ( $dE/dx$ ):** Measures the rate of specific ionization loss to identify particles via Bethe-Bloch curves at low momentum.

# Particle Identification (PID): Time-of-Flight vs. RICH

## Time-of-Flight (TOF) Systems

- Measures the time difference  $\Delta t$  between collision vertex and detector impact.
- **\*\*TOF Resolution:\*\*** High-precision Multigap Resistive Plate Chambers (MRPC) achieve resolutions of  $\sigma_t \sim 50\text{--}80$  ps.
- Separates pions, kaons, and protons at low-intermediate transverse momenta.

## Ring Imaging Cherenkov (RICH)

- Relies on Cherenkov radiation emitted when a particle's speed exceeds speed of light in medium:  $\beta > 1/n$ .
- Measures Cherenkov angle  $\theta_C$ :

$$\cos \theta_C = \frac{1}{n\beta}$$

- Projects rings onto photon detectors, separating high-momentum species.

# Energy Capture: Calorimeters & Muon Spectrometers

## Calorimetry Layers

- **\*\*Electromagnetic Calorimeter (EMCal):\*\*** Uses high-Z absorbers (e.g. Lead-Tungstate crystals) to trigger Bremsstrahlung showers, capturing all  $e^\pm$  and  $\gamma$  energy.
- **\*\*Hadronic Calorimeter (HCal):\*\*** Absorbs strongly interacting hadrons ( $\pi^\pm, K^\pm, p, n$ ) through inelastic nuclear interactions.

## Muon Spectrometers

- Muons are minimum ionizing particles (MIPs) and do not trigger hadronic or electromagnetic showers.
- They penetrate through thick iron shielding.
- Placed outermost to record clean muon tracks (crucial for vector meson decay  $J/\psi \rightarrow \mu^+ \mu^-$ ).

# Why AMPT? The Dynamical Multi-Phase Transport

## The Physics Motivation

- Detectors only record the asymptotic hadronic state after **kinetic freeze-out**.
- **AMPT (A Multi-Phase Transport)** is a dynamical Monte Carlo transport model that reconstructs the complete collision history:

Initial Geometry (HIJING)  $\rightarrow$  Parton Cascade (ZPC)  $\rightarrow$  Hadronization (Coalescence)  $\rightarrow$  Hadron Transport

## Course Outcomes for Students

By the end of this 14-day course, you will be able to:

- 1 Stream and parse raw simulated heavy-ion data.
- 2 Code relativistic kinematics and flow observables ( $v_2$ ) from scratch.
- 3 Build full reproducible data analysis pipelines.

# Relativistic Kinematics: Collinear Boosts (Sahoo pg. 115-116)

## Successive Boosts along X-axis

Consider three collinear inertial frames of reference  $S \xrightarrow{v} S' \xrightarrow{u} S''$  moving along the positive X-axis (Sahoo Fig. 5.6):

**\*\*Boost 1:  $S \rightarrow S'$  with velocity  $v$  ( $\beta_1 = v/c$ ):\*\***

$$x'^0 = \gamma_v(x^0 - \beta_1 x^1)$$

$$x'^1 = \gamma_v(x^1 - \beta_1 x^0)$$

where  $\gamma_v = \frac{1}{\sqrt{1-\beta_1^2}}$

**\*\*Boost 2:  $S' \rightarrow S''$  with velocity  $u$  ( $\beta_2 = u/c$ ):\*\***

$$x''^0 = \gamma_u(x'^0 - \beta_2 x'^1)$$

$$x''^1 = \gamma_u(x'^1 - \beta_2 x'^0)$$

where  $\gamma_u = \frac{1}{\sqrt{1-\beta_2^2}}$

# Relativistic Velocity Addition

## Combined Single Boost $S \rightarrow S''$ with velocity $w$

Substituting the coordinate equations from Boost 1 into Boost 2 yields the effective relativistic velocity boost factor (Sahoo Eq. 5.41):

$$w = \frac{v + u}{1 + \frac{vu}{c^2}}$$

And the total combined Lorentz factor becomes (Sahoo Eq. 5.39):

$$\gamma_w = \gamma_u \gamma_v \left( 1 + \frac{uv}{c^2} \right)$$

## Interactive Relativistic Demonstrations

Open in your browser:

- [animations/rapidity\\_boost.html](#) (Linear rapidity boosts & additivity)
- [animations/01\\_velocity\\_saturation\\_comparison.html](#) (Velocity saturation vs

# Rigorous Limits of Velocity Addition

## Universal Speed Limit

This non-linearity guarantees that the speed of light  $c$  is the absolute speed limit in the universe:

$$\text{If } v \rightarrow c \text{ or } u \rightarrow c \Rightarrow w \rightarrow c$$

## Classical vs. Relativistic Velocity Saturation

In Problem 1 of the hands-on lab, you will successively add collinear boosts of  $0.1c$  up to 20 times. - **\*\*Classical sum\*\*** grows linearly to  $2.0c$ . - **\*\*Relativistic sum\*\*** saturates at  $0.964c$ , asymptotically approaching  $1.0c$ , as shown in your saved plot `day01_problem1_velocity_rapidity.png`.

# Rapidity Definition (Sahoo pg. 117-118)

## Hyperbolic Transformations in Minkowski Space

Because velocity addition is highly non-linear, measuring spectrum shapes and calculating transformations across moving frames in colliders is mathematically complex. We define a new hyperbolic angle called **Rapidity** ( $y$ ):

$$y = \tanh^{-1}(\beta) = \frac{1}{2} \ln \left( \frac{1 + \beta}{1 - \beta} \right)$$

Hyperbolic mappings transform the velocity parameter:

$$\beta = \tanh(y), \quad \gamma = \cosh(y), \quad \gamma\beta = \sinh(y)$$

# Proof of Rapidity Additivity: Mathematical Derivation

## Hyperbolic Angle Transformations (Sahoo pg. 117-118)

Let  $y_1 = \tanh^{-1}(u/c)$  and  $y_2 = \tanh^{-1}(v/c)$  represent two collinear boosts (taking  $c = 1$ ):

$$w = \frac{u + v}{1 + uv} = \frac{\tanh y_1 + \tanh y_2}{1 + \tanh y_1 \tanh y_2}$$

## The Hyperbolic Tangent Identity

Recalling the standard hyperbolic tangent angle-addition identity:

$$\tanh(y_1 + y_2) = \frac{\tanh y_1 + \tanh y_2}{1 + \tanh y_1 \tanh y_2}$$

Which gives us the combined velocity parameter:

$$w = \tanh(y_1 + y_2)$$

# Rapidity Additivity: Physical Consequences

## Proving Linear Additivity (Sahoo Eq. 5.43)

Since the effective relativistic velocity boost is  $w = \tanh(y_{\text{tot}})$ , substituting our derivation yields:

$$\tanh(y_{\text{tot}}) = \tanh(y_1 + y_2) \quad \Rightarrow \quad y_{\text{tot}} = y_1 + y_2$$

This simple, elegant relation proves that collinear boosts combine linearly under rapidity!

## Takeaway: Natural Collider Spectroscopy Variable

\* **Velocity Addition** is highly non-linear, making cross-frame spectrum transformations mathematically complex. \* **Rapidity Addition** is linear. Boosts along the beam axis simply shift the rapidity distribution by a constant ( $y \rightarrow y + y_{\text{boost}}$ ), keeping the shape of the invariant yield distribution  $dN/dy$  perfectly invariant!

# Kinematics in the Center-of-Momentum System (Sahoo pg. 113-114)

## Asymmetric 2-Body Kinematics

Let  $p_1$  and  $p_2$  be the momentum 4-vectors of two colliding particles with masses  $m_1$  and  $m_2$ .  
Let  $P = p_1 + p_2$  be the total center-of-momentum 4-vector, with total invariant mass squared:

$$P^2 = (p_1 + p_2)^2 = M^2 = s$$

## Individual CMS Energies (Sahoo Eq. 5.29 & 5.30)

Transforming to the center-of-momentum frame ( $P = (\sqrt{s}, \vec{0})$ ) yields:

$$E_1^* = \frac{s + m_1^2 - m_2^2}{2\sqrt{s}}$$

$$E_2^* = \frac{s - m_1^2 + m_2^2}{2\sqrt{s}}$$

# CMS Momentum & Relative Velocity

## CMS Momentum $p^*$ (Sahoo Eq. 5.32)

By using the invariant mass relation  $E_1^{*2} - |\vec{p}^*|^2 = m_1^2$ , we solve for the exact center-of-momentum three-momentum magnitude:

$$|\vec{p}^*|^2 = \frac{s^2 - 2s(m_1^2 + m_2^2) + (m_1^2 - m_2^2)^2}{4s}$$

Note that  $E_1^* + E_2^* = \sqrt{s}$  and the particles move in opposite directions with equal momentum magnitude:  $\vec{p}_1^* = -\vec{p}_2^*$ .

## Relative CMS Velocity (Sahoo Eq. 5.33)

The relative CMS velocity parameter is calculated as:

$$v_{12}^* = \left( \frac{|\vec{p}^*|}{E_1^*} \right)^2$$

# Real Data: Symmetric vs. Asymmetric CMS Kinematics

## Computed Metrics from Problem 2

These exact kinematic formulas yield the following physical values for symmetric and asymmetric collider configurations:

- **Symmetric  $p + p$  at ALICE ( $\sqrt{s} = 13.6$  TeV):**

$$E_1^* = E_2^* = 6800.000 \text{ GeV}, \quad p^* = 6800.000 \text{ GeV}/c$$

- **Symmetric  $Au + Au$  at STAR ( $\sqrt{s_{NN}} = 200$  GeV):**

$$E_{\text{nucleon}}^* = 100.000 \text{ GeV}, \quad p^* = 99.996 \text{ GeV}/c$$

- **Asymmetric  $p + Pb$  at LHC ( $\sqrt{s_{NN}} = 8.16$  TeV):**

$$E_p^* = 4077.666 \text{ GeV}, \quad E_{Pb}^* = 4082.334 \text{ GeV}, \quad p^* = 4077.666 \text{ GeV}/c$$

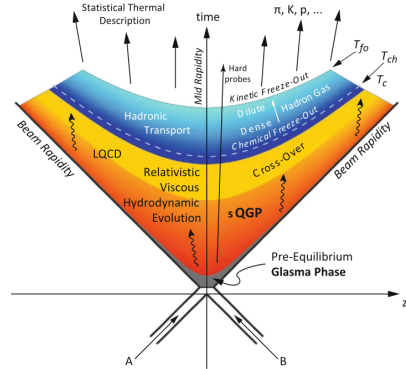
# Spacetime Evolution Stages of Heavy-Ion Collisions

## Dynamical Stages (Sahoo Fig. 5.7)

Relativistic collisions evolve through highly distinct physical epochs:

- 1 **\*\*Initial Collision (Glasma Phase):\*\*** Extreme gluon fields at  $t \approx 0$  fm/c.
- 2 **\*\*Pre-Equilibrium & Thermalization:\*\*** Rapid interaction creates a deconfined system.
- 3 **\*\*Viscous Relativistic Hydrodynamics:\*\*** Strongly coupled QGP expands as a fluid ( $0.5 < t < 10$  fm/c).
- 4 **\*\*Hadronization:\*\*** Fluid cools to a hadron gas at critical temperature  $T_c \approx 155$  MeV.
- 5 **\*\*Hadronic Gas Transport (ART):\*\*** Hadrons collide, scatter, and freeze out.

## Chronological Spacetime Diagram



**Fig. 5.7** Description of heavy-ion collisions in one space ( $z$ ) and one time ( $t$ ) dimension, with two nuclei colliding at ( $t = 0, z = 0$ ). The formation of a very hot and dense fireball is envisaged, which follows a complex and dynamical spacetime evolution involving different quanta of the system at different time scales. The fireball passes through a pre-equilibrium phase, followed by the formation of a QGP and a cross-over phase transition to a hadron gas. This transition is marked by a critical temperature,  $T_c$  after which the chemical freeze-out and kinetic freeze-out of the system happen at temperatures  $T_{fo}$  and  $T_{ch}$  respectively. Final state secondary particles free-stream after

# Coordinate Mappings: Light-Cone Momentum Variables

## Definitions (Sahoo pg. 119-120)

For ultra-relativistic collinear longitudinal processes, we define light-cone momentum variables:

$$p_+ = E + p_z, \quad p_- = E - p_z$$

Which are highly advantageous since they scale simply under longitudinal boosts:

$$p'_+ = e^{-y_{\text{boost}}} p_+, \quad p'_- = e^{y_{\text{boost}}} p_-$$

## Relation to Standard Rapidity

Using these variables, standard momentum rapidity  $y$  can be expressed as:

$$y = \frac{1}{2} \ln \left( \frac{p_+}{p_-} \right)$$

In Problem 4, you verify that this formulation matches standard rapidity with exact

# Coordinate Mappings: Spacetime Rapidity $\eta_s$

## Longitudinal Spacetime Scaling (Sahoo Eq. 5.44)

To represent the spatial coordinates  $(z, t)$  of particle freezeouts in an expanding QGP medium, we define the **\*\*Spacetime Rapidity\*\*** ( $\eta_s$ ):

$$\eta_s = \frac{1}{2} \ln \left( \frac{t+z}{t-z} \right)$$

Analogously, we define the proper lifetime  $\tau$ :

$$\tau = \sqrt{t^2 - z^2}$$

## Bjorken Longitudinal Scaling Model

In ultra-relativistic collisions, spatial position and momentum become highly correlated: particles with large forward velocities end up at large forward spatial coordinates.

# Real AMPT Correlation: Spacetime vs. Momentum Rapidity

## Validation on 27,617 Pion Tracks

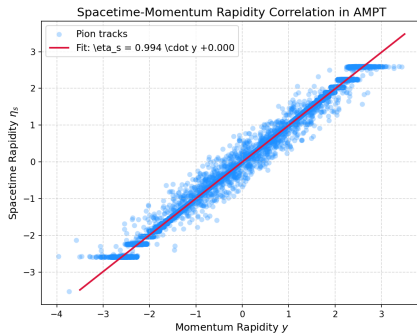
In Problem 5 of the laboratory, you map out the  $(y, \eta_s)$  scatter correlation for freezeout pions using real AMPT simulations.

- **Linear correlation  $r$ :** 0.9973
- **Fitted slope:** 0.9937

## Physical Conclusion

This near-unity slope and high correlation coefficient represent a direct validation of the **Bjorken scaling model** of relativistic longitudinal expansion!

## AMPT Spacetime-Momentum Correlation



# 3D Kinematics in Detector Planes

## Transverse & Longitudinal Decomposition (Sahoo Fig. 5.8)

A particle's momentum  $\vec{p}$  is decomposed relative to the beam axis (Z-axis) into: -

**\*\*Transverse Momentum ( $p_T$ ):\*\*** curvature in TPC:  $p_T = \sqrt{p_x^2 + p_y^2}$  - **\*\*Transverse Mass**

**( $m_T$ ):\*\***  $m_T = \sqrt{p_T^2 + m_0^2}$

**\*\*Azimuthal Angle  $\phi$  (Sahoo Eq. 5.50):\*\***

$$\phi = \tan^{-1} \left( \frac{p_y}{p_x} \right) \quad [-\pi, \pi]$$

**\*\*Polar Angle  $\theta$  (Sahoo Eq. 5.51):\*\***

$$\theta = \tan^{-1} \left( \frac{p_T}{p_z} \right) \quad [0, \pi]$$

# Hyperbolic Mappings to Rapidity

## Standard Rapidity $y$ in term of $m_T$ (Sahoo Eq. 5.54)

$$y = \frac{1}{2} \ln \left( \frac{E + p_z}{E - p_z} \right) = \ln \left( \frac{E + p_z}{m_T} \right)$$

Which highlights the hyperbolic boost rotation of coordinates:

$$E = m_T \cosh y, \quad p_z = m_T \sinh y$$

## Pseudorapidity $\eta$ (Sahoo Eq. 5.48)

For highly relativistic particles ( $E \gg m_0 \Rightarrow m_T \approx p_T$ ), standard rapidity simplifies to a purely geometric quantity called **\*\*Pseudorapidity\*\*** ( $\eta$ ), which depends only on polar angle  $\theta$ :

$$\eta = \tanh^{-1} \left( \frac{p_z}{p} \right) = -\ln \tan \left( \frac{\theta}{2} \right)$$

# Summary: Day 1 Hands-On Laboratory Exercises

## Objective: Build custom Relativistic & Coordinate Calculators

All exercises are framed in your Jupyter Notebook at `exercises/day01_hands_on.ipynb`:

- 1 **\*\*Problem 1:\*\*** Relativistic velocity addition vs. rapidity additivity (Plot saturation).
- 2 **\*\*Problem 2:\*\*** 2-body CMS energy mappings for symmetric ( $p + p$ ,  $Au + Au$ ) and asymmetric ( $p + Pb$ ) systems.
- 3 **\*\*Problem 3:\*\*** 3D Kinematics and coordinate decompositions (transverse variables and angles).
- 4 **\*\*Problem 4:\*\*** Light-cone momentum & rapidity on real AMPT data.
- 5 **\*\*Problem 5:\*\*** Spacetime rapidity mappings and correlation checks.